RIP ni sozlash

Configuring RIP

Asosiy atamalar

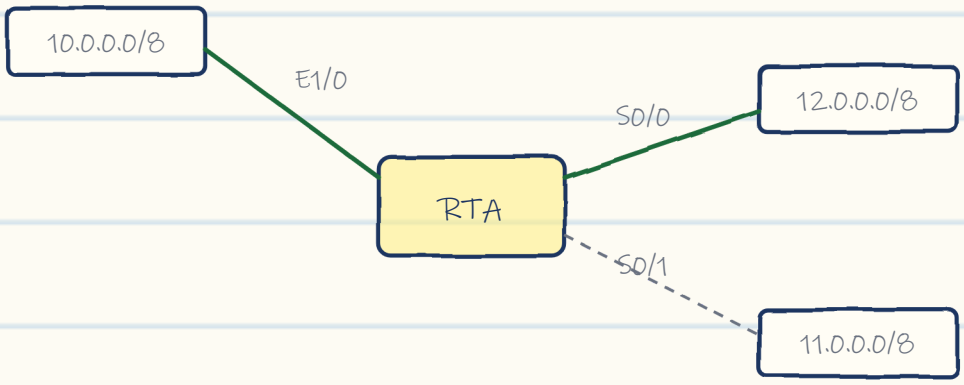
- network buyrug'i — qaysi interfeyslarda RIP ishlashini belgilaydi
- RIPv1 / RIPv2 — maskasiz / maskali, v2 — multicast 224.0.0.9
- Silent interface — eshitadi, lekin yangilanish yubormaydi
- Summary — avtomatik jamlash, undo bilan o'chiriladi
- Autentifikatsiya — yangilanishlarni parol/MD5 bilan himoyalash
- display rip — protokol holati va taymerlarini ko'rsatadi

1. Asosiy sozlash

[RTA]	rip [process-id]
[RTA-rip-1]	network network-address [wildcard-mask]

- process-id yozilmasa — sukut bo'yicha 1
- wildcard-mask — maskaning teskarisi; network 0.0.0.0 — hamma interfeys

network = qaysi interfeysda RIP ishlaydi



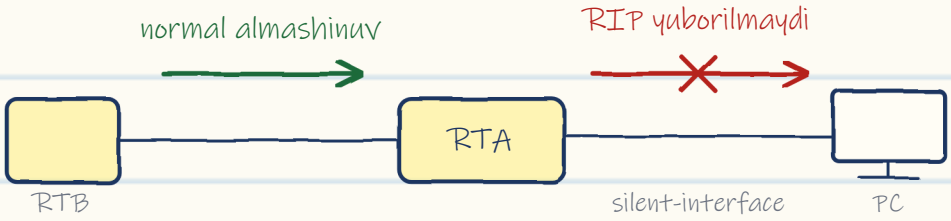
network 10.0.0.0 · network 12.0.0.0 → RIP yoqilgan

S0/1 e'lon qilinmagan → 11.0.0.0 tashqarida

1-rasm. network — ham qaysi direct marshrut e'lon qilinishini, ham qaysi interfeys paket olishini belgilaydi

2. Qo'shimcha sozlashlar

[RTA-rip-1]	silent-interface { all | interfeys }
[RTA-Eth1/0]	rip split-horizon
[RTA-Eth1/0]	rip poison-reverse

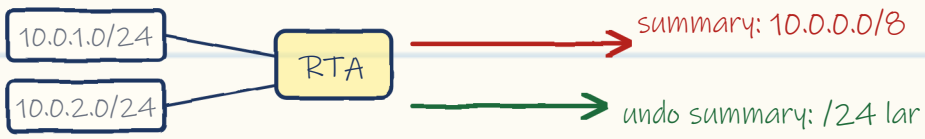


split horizon — sukut bo'yicha yoqilgan · poison reverse — o'chirilgan

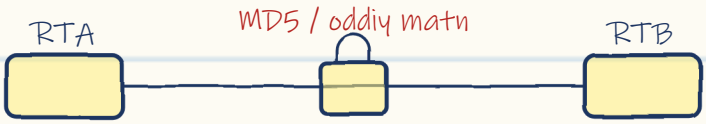
2-rasm. Jim interfeys yangilanish yubormaydi, lekin qabul qiladi

3. RIPv2 sozlash

[RTA-rip-1]	version { 1 | 2 }
[RTA-rip-1]	undo summary
[RTA-Ser2/0]	rip authentication-mode md5 rfc2453 plain PAROL



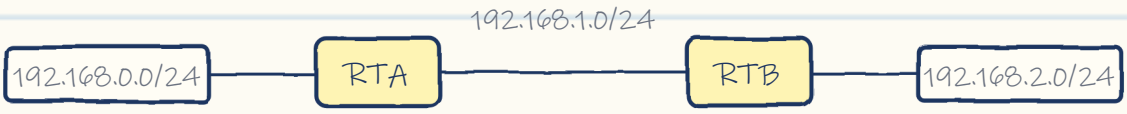
RIPV1 da avtosummarizatsiya o'chirilmaydi



parol: plain 1–16 belgi · cipher 33–53 belgi

3-rasm. undo summary — barcha subnetlar alohida e'lon qilinadi

4. RIPv1 misoli



hamma tarmoq tabiiy maskada → RIPv1 yetarli

RTA jadvali: 192.168.2.0/24 RIP 100 1 → 192.168.1.2

cost 1 = bir hop uzoqlikda

4-rasm. Ikkala routerda faqat rip + network yetarli

RTA	rip network 192.168.0.0 network 192.168.1.0
RTB	rip network 192.168.1.0 network 192.168.2.0

5. RIPv2 misoli



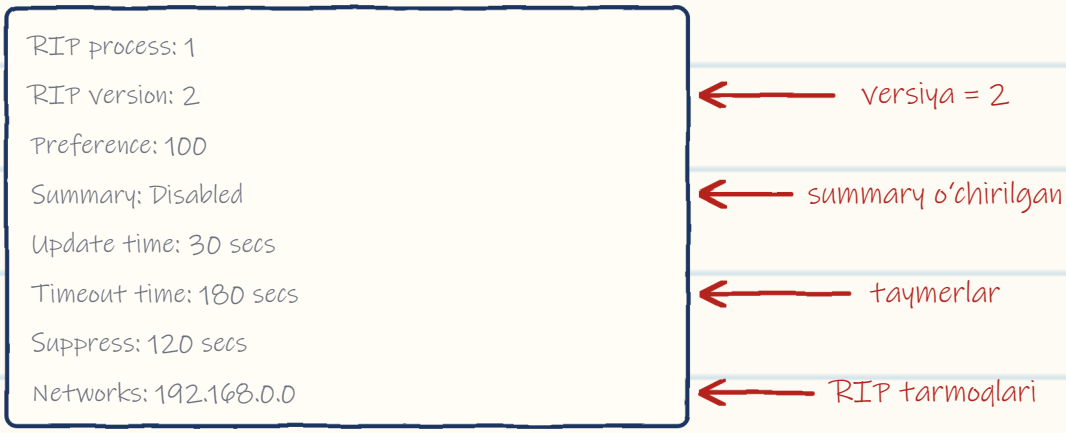
version 2 · undo summary · MD5

maska /30 va /24 — uzluksiz emas → RIPv2 kerak

5-rasm. RTA jadvaliga 10.0.2.0/24 RIP orqali tushadi

RTA · RTB	network 10.0.0.0 · network 192.168.0.0 version 2 · undo summary

6. Holatni ko‘rish



6-rasm. display rip — versiya, summarizatsiya, taymerlar va tarmoqlar

- debugging rip 1 packet — yuborilgan va olingan xabarlar
- chiqishda: versiya, broadcast yoki multicast, manzil va cost

Xulosa

- rip → network — RIP ni yoqishning ikki qadami
- network e’lon qilinmagan interfeys RIP dan tashqarida qoladi
- silent-interface — yubormaydi, qabul qiladi
- RIPv2: version 2 · undo summary · authentication-mode
- Tekshirish: display rip · display ip routing-table · debugging

Tekshirish

1. Sukutdagi process-id? — 1
2. Hamma interfeysda yoqish? — network 0.0.0.0
3. Avtosummarizatsiyani o‘chirish? — undo summary (faqat RIPv2)
4. Qaysi rejimda autentifikatsiya? — interfeys rejimida